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ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME

CONTEXT + EXACT CORE

Endogenic geomorphic activity includes tectonism, orogeny, epeirogeny, earthquakes and vulcanism generated by mantle heat, gravity, isostatic adjustment and lithospheric-plate interaction.

MECHANISM / ARGUMENT

plate motion concentrates compression, tension, shear and melting

CONSEQUENCE / CONTRAST

convergent belts host arcs + deep quakes; transforms host quakes without major vulcanism

UPSC TRAP / ANSWER USE

plate setting is the regional driver, not a substitute for rupture/magma mechanics

ANSWER LINE: Earthquakes and vulcanism are surface expressions of an internally powered, mobile lithosphere: plate motion supplies the regional setting, while fault rupture or magma ascent supplies the immediate mechanism.

02

STRESS, STRAIN, ELASTIC REBOUND AND FAULTS

CONTEXT + EXACT CORE

An earthquake commonly results from brittle failure and coseismic slip on a fault after tectonic stress produces elastic strain; elastic rebound is the partial recovery of deformed rock blocks after rupture.

MECHANISM / ARGUMENT

load - lock - strain accumulation - rupture/slip - elastic rebound

CONSEQUENCE / CONTRAST

tension-normal | compression-reverse/thrust | shear-strike-slip

UPSC TRAP / ANSWER USE

a mapped fault indicates a source possibility, not the time of the next earthquake

ANSWER LINE: The earthquake is sudden, but its preparation is usually slow: stress loads a locked fault, elastic strain accumulates, rupture begins when resistance is exceeded, and released energy radiates as seismic waves.

03

EARTHQUAKE TERMINOLOGY, CAUSES AND CLASSIFICATIONS

CONTEXT + EXACT CORE

An earthquake is the transient vibration of the ground caused by seismic-wave radiation from a sudden source, most commonly fault rupture but also volcanic, collapse, reservoir-associated, induced or explosion sources.

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MECHANISM / ARGUMENT

source class + focal depth + tectonic setting explain where and how shaking begins

CONSEQUENCE / CONTRAST

shallow often damages locally; deep slab events may be widely felt but differently expressed

UPSC TRAP / ANSWER USE

depth influences shaking, but vulnerability determines whether hazard becomes disaster

ANSWER LINE: Earthquake classification must distinguish the source mechanism, focal depth and tectonic setting; “tectonic earthquake” is the dominant class, not the only possible source of ground vibration.

04

FOCUS, EPICENTRE, SEISMOGRAPHS AND LOCATING AN EVENT

CONTEXT + EXACT CORE

The hypocentre/focus is the calculated subsurface origin of seismic radiation, while the epicentre is its vertical surface projection; a seismometer senses motion and a seismograph/seismic station records a time series called a seismogram.

MECHANISM / ARGUMENT

one station gives distance - three or more station circles intersect - epicentre is estimated

CONSEQUENCE / CONTRAST

epicentre locates the surface projection; worst damage may occur elsewhere

UPSC TRAP / ANSWER USE

a single station cannot supply both distance and unique direction

ANSWER LINE: Focus and epicentre identify source geometry, not damage ranking: the worst-hit locality may lie away from the epicentre because rupture directivity, basin amplification and vulnerable construction modify shaking.

05

P, S, LOVE AND RAYLEIGH WAVES

CONTEXT + EXACT CORE

Body waves comprise compressional P waves and shear S waves travelling through the Earth; surface waves comprise Love and Rayleigh modes guided near the surface and commonly producing large damaging motions.

MECHANISM / ARGUMENT

elastic properties control velocity, particle motion and permitted medium

CONSEQUENCE / CONTRAST

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CONTEXT + EXACT CORE

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MECHANISM / ARGUMENT

elastic properties control velocity, particle motion and permitted medium

CONSEQUENCE / CONTRAST

P arrives first and crosses liquids; S is stopped by liquid; surface waves strongly affect structures

UPSC TRAP / ANSWER USE

transverse S motion is not the same as horizontal-only Love motion

ANSWER LINE: P and S waves diagnose the Earth's interior because their velocity and permitted media differ, whereas Love and Rayleigh waves dominate the near-surface damage problem because their amplitudes and durations can be large.

REFRACTION, DISCONTINUITIES AND SHADOW ZONES

CONTEXT + EXACT CORE

Seismic shadow zones are angular regions where direct wave phases are absent because P waves are strongly refracted at the core–mantle boundary and S waves cannot propagate through the liquid outer core.

MECHANISM / ARGUMENT

P bends strongly through the core; S cannot cross a liquid outer core

CONSEQUENCE / CONTRAST

P later reappears; direct S remains absent on the far side

UPSC TRAP / ANSWER USE

S-wave absence proves liquid outer core, not an entirely liquid core

ANSWER LINE: The Earth's deep interior is known by wave behaviour rather than direct observation: P-wave refraction and S-wave extinction reveal sharp discontinuities and the liquid state of the outer core.

07

MAGNITUDE, INTENSITY AND LOGARITHMIC ENERGY

CONTEXT + EXACT CORE

Magnitude is an instrument-derived event parameter—large events are commonly reported using moment magnitude M_w —whereas intensity scales such as MMI or MSK classify observed shaking and effects at individual locations.

MECHANISM / ARGUMENT

seismic moment estimates source size; observations classify place-specific shaking

CONSEQUENCE / CONTRAST

one magnitude but multiple intensities and unequal losses

UPSC TRAP / ANSWER USE

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MECHANISM / ARGUMENT

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CONSEQUENCE / CONTRAST

one magnitude but multiple intensities and unequal losses

UPSC TRAP / ANSWER USE

never convert a magnitude directly into a fixed damage category

ANSWER LINE: An earthquake has one estimated magnitude but many intensities, because rupture size is an event property while local effects vary with distance, depth, path, ground and buildings.

GLOBAL EARTHQUAKE DISTRIBUTION AND PLATE MARGINS

CONTEXT + EXACT CORE

Earthquake distribution maps delineate subduction Wadati–Benioff zones, mid-ocean ridges, transform faults, continental collision belts and a smaller population of intraplate sources.

MECHANISM / ARGUMENT

subduction gives shallow-deep seismicity; ridges/transforms give shallow seismicity; inherited faults reactivate inside plates

CONSEQUENCE / CONTRAST

seismic belts overlap volcano belts strongly but not perfectly

UPSC TRAP / ANSWER USE

collision Himalaya is seismic without being a typical oceanic-subduction volcanic arc

ANSWER LINE: The global map of seismicity is a map of plate interaction: shallow events outline almost every boundary, while inclined shallow-to-deep zones uniquely reveal subducting slabs.

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SECONDARY HAZARDS, SITE RESPONSE AND THE RISK EQUATION

CONTEXT + EXACT CORE

Earthquake risk is the potential loss arising from seismic hazard interacting with exposure, physical/social vulnerability and coping capacity; cascading hazards include liquefaction, slope failure, fire, surface rupture, dam failure and tsunami.

MECHANISM / ARGUMENT

source - path - site - building/network response - cascade

CONSEQUENCE / CONTRAST

equal magnitude does not produce equal loss across sites or societies

UPSC TRAP / ANSWER USE

do not use earthquake, hazard, risk and disaster as synonyms

ANSWER LINE: A single earthquake with a fixed magnitude can produce different levels of shaking at different locations, due to distance, depth, path, ground conditions and site-specific factors.

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UPSC TRAP / ANSWER USE

do not use earthquake, hazard, risk and disaster as synonyms

ANSWER LINE: An earthquake is a hazard event; it becomes a disaster when shaking and cascades strike exposed, vulnerable systems whose coping capacity is exceeded.

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TSUNAMI GENERATION, PROPAGATION AND COASTAL IMPACT

CONTEXT + EXACT CORE

Tsunamis are long-period gravity waves most commonly produced by rapid vertical sea-floor displacement during submarine megathrust earthquakes, but also by volcanic collapse/eruption, submarine landslides or rare impacts.

MECHANISM / ARGUMENT

submarine rupture - deep-ocean propagation - shallow-water amplification - coastal flooding

CONSEQUENCE / CONTRAST

low wave offshore can become destructive onshore; first wave need not be largest

UPSC TRAP / ANSWER USE

tsunami is a seismic/gravity wave train, not a tidal wave

ANSWER LINE: A tsunami is not a wind wave or tide: it begins with sudden water-column displacement, crosses deep ocean with low height and high speed, and grows destructively as shoaling compresses the wave near the coast.

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VOLCANO ANATOMY, MAGMA, LAVA, SILICA, GAS AND VISCOSITY

CONTEXT + EXACT CORE

Vulcanism comprises magma generation, ascent, intrusion and extrusion; magma is molten or partially molten subsurface material containing crystals and volatiles, while lava is magma erupted at the surface.

MECHANISM / ARGUMENT

silica + lower temperature + crystals - viscosity - trapped gas - fragmentation

CONSEQUENCE / CONTRAST

basaltic magma tends effusive; silicic magma tends explosive

UPSC TRAP / ANSWER USE

volcano shape is an outcome; magma rheology is the mechanism

ANSWER LINE: Eruption style is controlled less by the size of a cone than by magma rheology: silica-rich, cooler and gas-charged magma is commonly more viscous and explosive, whereas hotter basaltic magma generally allows easier degassing and effusive flow.

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INTRUSIVE AND EXTRUSIVE FORMS; VOLCANIC PRODUCTS

CONTEXT + EXACT CORE

Intrusive igneous bodies crystallise below the surface in concordant or discordant forms, whereas extrusive vulcanism constructs lava plains, cones, domes and pyroclastic deposits from erupted products.

MECHANISM / ARGUMENT

emplacement geometry forms intrusions; fragmentation and flow form extrusive products

CONSEQUENCE / CONTRAST

dyke cuts layers, sill follows layers; pyroclastic current is hot gas-debris, lahar is water-debris

UPSC TRAP / ANSWER USE

ash, sulphur compounds and nitrogen species are eruption products, not only lava

ANSWER LINE: Intrusive forms record the geometry of magma emplacement, while extrusive forms record the interaction of lava viscosity, gas, eruption rate, relief and erosion at the surface.

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ERUPTION STYLES, VOLCANO TYPES AND ACTIVITY STATUS

CONTEXT + EXACT CORE

Eruption-style classifications arrange behaviour from sustained effusive fissure or fountain activity to intermittent explosions and high-column Plinian/Peleian activity; morphology includes shield, composite, cinder, dome and caldera systems.

MECHANISM / ARGUMENT

easy degassing - effusion; trapped gas + viscous magma - fragmentation/explosion

CONSEQUENCE / CONTRAST

style, landform and activity status are independent classification axes

UPSC TRAP / ANSWER USE

absence of a recent historical eruption does not by itself prove extinction

ANSWER LINE: Eruption style, volcano shape and activity status are three separate classifications: fluid lava can build a shield, alternating lava and tephra can build a stratovolcano, and neither shape alone determines whether a volcano is extinct.

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PLATE-SETTING VOLCANISM, RING OF FIRE AND MID-OCEAN RIDGES

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CONTEXT + EXACT CORE

Boundary volcanism results from decompression melting at divergent margins and flux melting above subduction slabs, producing contrasting basaltic ridge/rift systems and commonly explosive volcanic arcs.

MECHANISM / ARGUMENT

divergence lowers pressure; subduction adds volatiles to the mantle wedge

CONSEQUENCE / CONTRAST

ridges yield basaltic crust; arcs yield mixed, often explosive stratovolcanism

UPSC TRAP / ANSWER USE

Ring of Fire means a geophysical system, not a perfect geometric circle

ANSWER LINE: Divergent volcanism is generated mainly by pressure release in upwelling mantle, whereas subduction volcanism is promoted by slab-derived volatiles that lower the melting temperature of the mantle wedge.

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MANTLE PLUMES, HOT SPOTS AND FLOOD BASALTS

CONTEXT + EXACT CORE

In the plume hypothesis, a buoyant thermal anomaly rises through the mantle; a broad plume head may produce a large igneous province, while a narrower persistent tail forms a hot spot and an age-progressive volcanic chain as the plate moves overhead.

MECHANISM / ARGUMENT

head reaches lithosphere - voluminous fissure basalt; moving plate over tail - volcanic chain

CONSEQUENCE / CONTRAST

Deccan plateau differs from an arc cone; hot-spot trail records plate motion

UPSC TRAP / ANSWER USE

plume hypothesis is a standard but qualified explanation, not a settled universal cause

ANSWER LINE: Mantle plumes complement plate-boundary theory by explaining selected intraplate volcanic chains and flood-basalt provinces, while hot-spot age progression provides an independent record of plate motion.

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CONSTRUCTIVE, DESTRUCTIVE, CLIMATIC AND ENVIRONMENTAL EFFECTS

CONTEXT + EXACT CORE

Volcanic impacts operate through lava, tephra, density currents, lahars, gases and aerosols at event scales, and through crustal construction, nutrient release, hydrothermal mineralisation and atmospheric forcing at longer scales.

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MECHANISM / ARGUMENT

product + height + wind + relief + water control the impact footprint

CONSEQUENCE / CONTRAST

short-term destruction/cooling can coexist with long-term land, soil and resource benefits

UPSC TRAP / ANSWER USE

ash is not the principal long-lived global cooling agent; stratospheric sulphate aerosol is

ANSWER LINE: Volcanism is geographically double-edged: episodic high-intensity hazards coexist with persistent benefits such as fertile weathered material, geothermal energy, mineralisation and new crust.

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INDIA'S TECTONIC FRAMEWORK AND REGIONAL SEISMIC RISK

CONTEXT + EXACT CORE

India's seismicity reflects Indian–Eurasian convergence, Indo-Burmese deformation, Sunda–Andaman subduction and reactivation of inherited intraplate structures within the Peninsular shield.

MECHANISM / ARGUMENT

collision/subduction drive boundary seismicity; old faults can reactivate inside the plate

CONSEQUENCE / CONTRAST

Himalaya/Andaman are boundary systems; Kachchh/Latur/Koyna correct the “stable means aseismic” error

UPSC TRAP / ANSWER USE

Indo-Gangetic alluvium may amplify Himalayan shaking far from the mountain source

ANSWER LINE: India does not possess one earthquake belt but a mosaic of collision, subduction, syntaxis, rift-reactivation and intraplate settings, each requiring a different source explanation and risk lens.

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INDIA SEISMIC ZONATION, MICROZONATION AND CLASSIFICATION CAUTION

CONTEXT + EXACT CORE

Macrozonation assigns broad design-hazard classes using national-scale seismic evidence, while microzonation integrates geology, seismotectonics, ground motion, soil and site response to produce finer GIS-based hazard or risk products.

MECHANISM / ARGUMENT

national evidence defines broad design zones; local geology/soil/groundwater refine site response

CONSEQUENCE / CONTRAST

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MECHANISM / ARGUMENT

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CONSEQUENCE / CONTRAST

zone map guides design but neither macro- nor microzonation predicts event time

UPSC TRAP / ANSWER USE

disclose the 2025/2026 official inconsistency instead of asserting an unverified settled Zone VI code

ANSWER LINE: Zonation is a design and planning input, not an earthquake prediction: macrozones generalise regional hazard, while microzonation resolves the local soil, basin and liquefaction contrasts that determine neighbourhood-level shaking.

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INSTITUTIONS, RESILIENT CONSTRUCTION, EARLY WARNING LIMITS AND PREPAREDNESS

CONTEXT + EXACT CORE

Seismic risk reduction combines monitoring, hazard assessment, code-compliant ductile design, non-structural safety, retrofitting, land-use control, lifeline continuity, drills, public education and rapid warning/response systems.

MECHANISM / ARGUMENT

monitor/map - design/enforce - retrofit lifelines - secure non-structural elements - drill/communicate

CONSEQUENCE / CONTRAST

early warning may provide seconds after rupture; prediction claims event before rupture

UPSC TRAP / ANSWER USE

codes on paper do not reduce vulnerability without local adoption, supervision and compliance

ANSWER LINE: Since deterministic earthquake prediction is unavailable, public policy must shift from forecasting the event to reducing the consequences through safer sites, ductile structures, retrofitted lifelines and practised community action.